

Digital Twins in the Future Design of Digital Therapeutics

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ABSTRACT

This paper contributes a methodological framework for the design of digital therapeutics for mental health which leverages on state-of-the-art knowledge on digital twins and virtual coaching solutions to realize more effective and AI-powered digital health interventions. The paper discusses how the deployment of digital twins as computational models for predicting patient's health condition can be combined with human-centered distributed conversational modeling techniques in designing AI powered mental health interventions with key stakeholders. This approach helps to address main challenges in the design of AI-powered digital therapeutics, such as optimizing design efforts and resources, ensuring ethics soundness and transparency of decisions related to the digital treatment, as well as informing the pre-clinical validation of future digital therapeutics.

CCS CONCEPTS

• Ubiquitous and mobile computing design and evaluation methods;

KEYWORDS

Digital Twins, Mental Health interventions, Digital Therapeutics, Virtual coaching solutions, Co-design, Socio-technical systems

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1 INTRODUCTION

A digital twin is a simulation technology that delivers digital health insights while also allowing prediction and recommendation within a feedback loop [1]. In the context of digital health and mental health in particular, a digital twin (DT) can be conceived as a computerized model of a person's psychological and emotional state,

created using data collected from various sources, such as wearable sensors, social media, and electronic health records. The use of DTs for mental health has the potential to revolutionize the way we approach mental health care. By creating a DT of a patient, mental health professionals can gain a more comprehensive understanding of the patient's needs and tailor their treatment accordingly [2]. For example, a DT could support mental health professionals in performing personalized stress prediction and monitoring [3] for considering and adapting different treatment options during the design and pre-clinical testing of digital stress management interventions. The use of DTs in mental health is still a relatively new concept, and there is limited research on its effectiveness. However, so far early studies and pilot projects in digital interventions for prevention and care of unipolar or bipolar disorder [4], as well as stress management [5-6], have shown promising results.

In this paper we discuss a methodological framework leveraging state-of-the-art knowledge on DTs for precision mental health and virtual coaching mHealth behavioral interventions to design more effective evidence-based digital therapeutics (DTx) powered by Artificial Intelligence (AI).

In the following, we first introduce key challenges in the design of future DTx for mental health, we then discuss a methodological framework combining DTs models, Human-centered Distributed Conversational Modeling (HDCM) techniques and AI as key components informing the design of evidence-based, ethically sound DTx treatments for mental health with key stakeholders. In section 4 we elaborate on the use of this framework in the socio-technical validation of DTx and in section 5 we conclude by summarizing the main open issues and benefits of our method for the design of more effective DTx for mental health.

2 CHALLENGES IN DESIGNING FUTURE DTX

Virtual coaches or chatbots that can communicate with the patient have been used as digital excipients to facilitate the delivery and intake of active ingredients of an evidence-based treatment [7], e.g., Cognitive Behavioral Therapy (CBT) for mental health, with the benefit of scaling up behavioral treatment to the target users by lowering barriers to patients' access to treatment. Although in recent years systematic frameworks for optimizing efforts and resources in the development of evidence-based behavioral treatments for prevention and care have been provided, such as the ORBIT model (Fig.1) in the field of chronic diseases [8], there is still room for improving and optimizing the methodologies leading to an effective design and preliminary testing of DTx (Phase I and Phase II of the ORBIT model).

An essential step for moving for example from Phase I (Design) to Phase II (Preliminary testing) of the ORBIT process is the confidence of the investigator that the treatment package is complete, ethically sound, that it includes essential components offered in an

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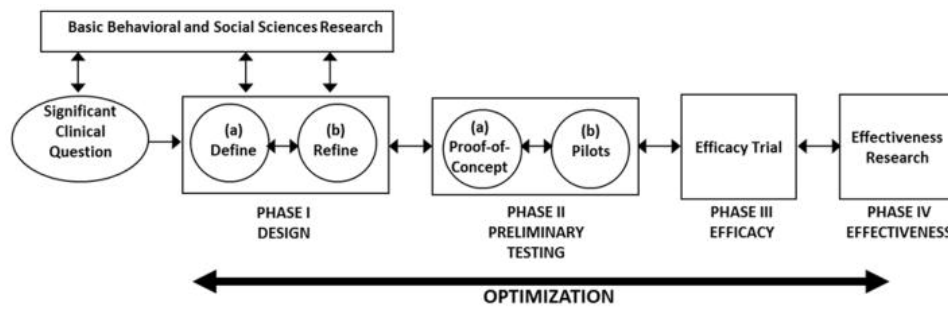


Figure 1: The ORBIT Model

efficient way, is safe, engaging, and acceptable to the population of interest and promises to have a clinically significant benefit on the behavioral risk factor. Only when this is achieved the treatment protocol can undergo a proof-of-concept evaluation by reducing the risk of investing considerable effort and resources in the evaluation of a sub-optimal treatment protocol.

3 LEVERAGING ON DTS MODELS, AI AND HDC TECHNIQUES IN THE DESIGN OF DTX

Our methodological framework calls for addressing the initial design and refinement of DTx by involving key stakeholders in the mental health domain as co-designers of the treatment protocol in a collaborative simulation and modeling environment. The design space can be populated with DTs of relevant patient profiles from the target user group, helping to predict and monitor relevant patient's condition (such as stress condition) and possible health trajectories and risk factors that can be formalized into a computational model for simulation purposes. The simulation environment needs also to exploit AI techniques, such as data mining and machine learning (ML) to build the prediction capabilities of the DT by using a dynamic ML process [9]. For building the DT coaching capabilities the mental health experts can leverage on the deployment of Human-centered Distributed Conversational Modeling (HDCM, [10]) techniques to iteratively design and refine the DTx protocol during the pre-clinical phase, in collaboration with relevant groups of stakeholders, such as healthcare professionals (domain experts), healthcare trainees (domain novices), healthy volunteers or patients' representatives.

By deploying this HDCM approach the design and refinement of a virtual coaching model and protocol (corpus of dialogues required to delivering a DTx treatment) can rely on an iterative asynchronous collaborative process following these main phases:

a) A domain expert seeds the virtual coaching model with a structured dialogical script/protocol of what a virtual coach (e.g., chatbot providing CBT stress management treatment) will say to a specific patient during a treatment and what the patient may say back.

b) Multiple novices, healthy volunteers or patients have a typed conversation with the virtual coach by playing the role of the target patients' group. The co-design process collects new stimuli when the virtual coach does not have a response, and when it responds incorrectly.

c) A domain expert enters responses to which the virtual coach could not respond, or to which the virtual coach responded incorrectly.

Phases b and c are repeated until an acceptable accuracy is reached, determined by the domain expert.

d) Domain experts and novices can have meta-discussions on the different sessions of the virtual coaching protocol to further support its refinement and overall compliance with the treatment objectives and the needs of the target patients' profiles. Stakeholders are provided with dedicated online collaboration tools supporting the iterative asynchronous design and simulation process.

This methodological approach can address several challenges in the creation of effective and comprehensive corpora to develop and train virtual coaches for DTx. First, the elicitation of a sufficient number of questions and responses between the virtual coach and the patient to enable generalizability and flexibility of the treatment deployment to different types of patients' needs and profiles. Second, overcoming typical barriers in the creation of conversation datasets, such as the availability of patients interacting with prototyped virtual coaches or chatbots, the time required for the data collection, the ethics, and legal requirements among others. Third, overcoming difficulties and costs required by knowledge engineers to acquire information from domain experts during the design and refinement of the conversation protocol, a process which might be even worsened by the differing backgrounds. Therefore, designing a DTx in a collaborative simulation environment informed by DTs, AI and HDCM techniques can represent a more effective and safer approach for prototyping, adapting, and optimizing a treatment package in the pre-clinical phase before its large-scale deployment with patients.

Furthermore, from an ethical perspective our methodological framework would support an easier definition and understanding of the mediating role that a virtual coaching intervention should play in a therapeutic setting. The stakeholders' discussions and meta-analyses conducted in phase d may help to better define the virtual coach capabilities and objectives, by ensuring for example that they support psycho-educational goals and properly mediate CBT techniques and skills, instead of trying to mimic human conversation with the risk of leading users to false beliefs and wrong expectations towards the digital treatment [11]. In this sense our simulation environment may help to explore and better clarify the role of DTx as novel forms of epistemic exchange, as well as the

normative conditions under which patients' health and well-being can benefit most.

4 SOCIO-TECHNICAL APPROACH TO DTx DESIGN, IMPLEMENTATION AND VALIDATION

Our approach is inspired by a socio-technical perspective that considers technological artifacts as part of a heterogeneous ecology of work practices, knowledge, and organizational processes. In this perspective, we propose to consider DTx as elements that will have to be part of a complex system to which they will have to partly adapt and partly contribute to modify. In this respect, the fate of DTx thus depends not only on their intrinsic therapeutic effectiveness and efficiency but also on their integration within the healthcare setting and the system of health professions.

Adopting an ecological approach means, firstly, recognising that the possibility of creating and evolving DTs requires the alignment and cooperation of various actors and stakeholders. If we take up the preliminary definition given in the introduction, the creation of a DT needs data from different sources (e.g., mobile and wearable sensors, social media, electronic health records), and only by guaranteeing long-term access does the alignment of the DT to the user's profile. In this respect, the design of the DT must provide ways to align the different data providers by offering 'rewards' that ensure their involvement.

Secondly, an ecological perspective points to the desirability of adopting participatory design tools at all stages from requirements definition to validation in the context of use. Alongside pre-clinical and clinical validation, which is closely linked to the appropriateness of the content offered and the interactions between the patient and the technology, DTx calls into question the need to rethink organizational processes, the training of healthcare workers, the relationship between doctors and patients, and so on. Participatory design techniques make it possible not only to define the scripts of the interaction, but also to define the requirements needed to incorporate the new technology within organizational processes.

Thirdly, finally, our approach considers DTx based on DTs and AI as one of the treatment options of a toolbox available to healthcare professionals in a stepped care logic. In other words, we imagine that the use of conversational agents to deliver DTx should be conceived as one of the low human labor-intensive resources that can stand alongside other traditional therapies (e.g., pharmacological, group therapy, face-to-face therapy) in a logic that allows each patient to be offered the most appropriate and least costly response at each moment of his or her disease trajectory. Participatory design techniques, in our perspective, will allow to go beyond the design of the individual DTx to embrace the design of the clinical pathway for each condition of interest.

5 CONCLUDING REMARKS

Although the validation of DTx for mental health is still a long way off, it is important to flank the strictly engineering research with a reflection on the perspectives that the introduction of these technologies might have on the health professions, on organizational structures, and, at a higher level, on the policies needed to ensure their success. From a strictly professional point of view by analogy

with what has happened in other sectors, an educated guess is that the profile required of health professionals will be increasingly linked to the analysis of data or even the emergence of new specific professional figures. The introduction of DTs in healthcare also opens the door to the virtualization of organizational practices. As Croatti and colleagues note "digital twins could be the digital counterpart not only for physical computational assets (e.g., vital signs monitors, diagnostic machinery, surgery rooms, etc.) but also for care processes" [12].

In more general terms, however, the greatest challenge appears to be that of reconfiguring the health service (or some of its articulations) to accommodate a perspective that radically changes the relationship between doctors and patients. The predictive possibilities envisioned by DTs and the possibility of constructing twins for millions of people from a very young age could pave the way to a healthcare model based on prediction rather than on the acute event requiring a significant rethinking of healthcare service provision [13].

This paper intends to contribute initial thoughts to the debate addressing main design, ethical and validation concerns in the future development and adoption of DTs and AI-powered DTx for improving patients' health and well-being. These challenges regard not only the use of artificial intelligence for predictive purposes in healthcare, but also the impact that DTs and DTx will have in public health organizations involved in the future exploitation and management of these digital assets.

The recent advances in artificial intelligence and DTs promise to improve the quality of care by patients' empowerment, supporting patients to implement concepts and techniques (such as CBT) in everyday life, enabling the personalization of treatments and an earlier identification of health problems due to digital phenotyping [14]. However, the design of effective DTx brings also unprecedented challenges and many open questions that need to be inquired to better understand their impact on individuals and society in the long-term perspective.

The methodological framework discussed in this paper, which is currently deployed for the design of stress management interventions to be delivered during pregnancy, can help to better address and analyze these open questions from an applied research perspective. It can also inform the different phases of design and validation of DTx by optimizing the domain knowledge available and the specification of the intervention objectives and requirements in a more transparent, comprehensive, and effective way. By sharing our design approach with other researchers and practitioners in the DTx field we aim at identifying also possible synergies in the future development and adoption of AI-based simulation environments for DTx, as well as for the validation of DTx in different use case scenarios in a multidisciplinary and multi-centric perspective.

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